Decentralised Single-Index Bandits

Learning an unknown reward function without a server: findings so far

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Recap: single-index bandits

Model

Each round an agent sees K arms $x_{t,a} \in \mathbb{R}^d$, pulls one, and observes

$$\boxed{y_t = f(\langle x_t, \theta_* \rangle) + \eta_t}$$

θ_* and f unknown, f possibly non-monotone.

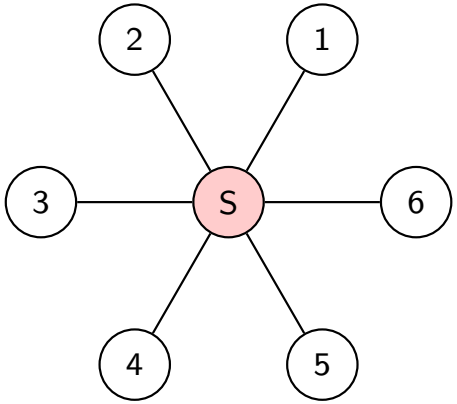
ZoomSIB-UCB (Dey, Bhore & Ghosh 2026): optimal $\tilde{\Theta}(T^{2/3})$

- **Phase 1:** explore; estimate θ_* with the Stein estimator $\hat{v} = \frac{1}{T_0} \sum_t y_t S(x_t) \propto \theta_*$, a *sample mean*.
- **Phase 2:** project $z = x^\top \hat{\theta}$, cut the line into bins, run *sleeping* UCB over the bins available each round. State: (n_j, S_j) per bin.

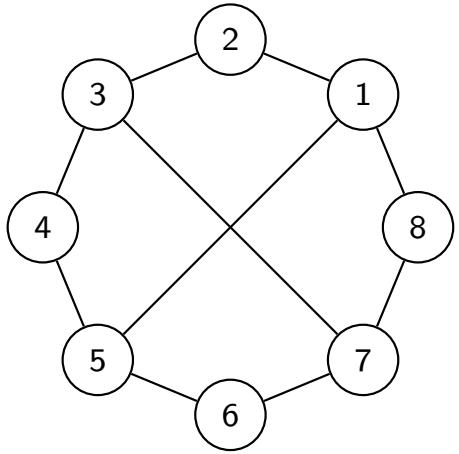
Our question

N agents share θ_* and f but **no server**: they talk only to graph neighbours.

With a server vs without a server



Server (star)
one coordinator sees all statistics



Serverless (graph G)
diameter D, spectral gap $1 - \lambda$

Why serverless?

No trusted coordinator, or no backbone: sensor networks, robot/drone swarms, vehicular networks, peer-to-peer recommendation, consortia that do not share raw data.

Network regret $R_T(N) = \sum_{i,t} \left[f(\langle x^*_{i,t}, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle) \right]$

Our algorithm: Dec-ZoomSIB-Relay

Phase 1: exact pooling without a server

Stein sums are **sufficient statistics**: spanning-tree aggregation (up and back down a BFS tree, $2\cdot$depth rounds) gives *every* agent the exact centralised $\hat{\theta}$.

Freeze: one shared grid

All agents freeze the same $\hat{\theta}$; the window W comes from max-consensus over D rounds.

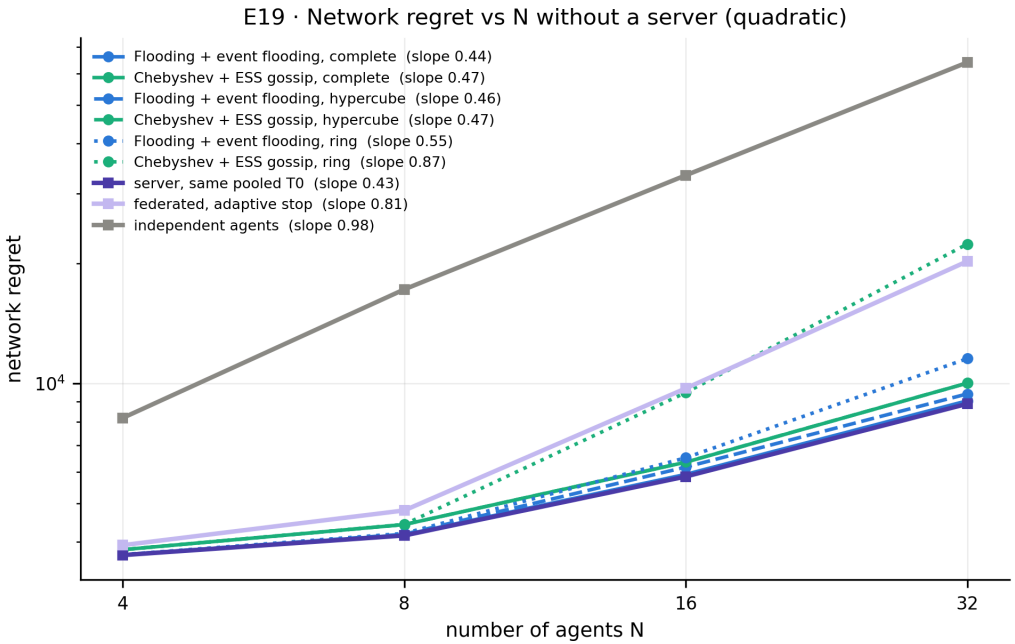
Phase 2: event-triggered flooding of bin tables

Each agent stores the freshest copy of every agent's own (n_j, S_j) table, relayed one hop per round. An agent republishes only when its unpublished count in a bin exceeds $\gamma\times$ what it currently sees.

Design principle

Relay exact, sparse updates instead of averaging.

Finding 1: collaboration pays without a server



E19, quadratic link, fixed pooled Phase-1 size

Network regret $\propto N^{\alpha}$, $N = 4 \ldots 32$:

	α
server	0.43
flooding, complete	0.44
flooding, hypercube ($D = \log N$)	0.46
flooding, ring ($D = N/2$)	0.55
independent agents	0.98

At $N = 32$: hypercube within **6%** of the server, ring within **30%**.

Mixing vs relaying: building a network sum

Every agent needs network totals: $\sum_i b_i$ (Phase 1) and n_j, S_j per bin (Phase 2). Toy example: ring of 4 agents with values $(4, 0, 0, 0)$; true average $= 1$.

Mixing (consensus / gossip)

Each round: $x_i \leftarrow \sum_j P_{ij} x_j$ (average with neighbours)

round	a_1	a_2	a_3	a_4
0	4	0	0	0
1	1.33	1.33	0	1.33
2	1.33	0.89	0.89	0.89
3	1.04	1.04	0.89	1.04

$\to 1$ but **never exact**; speed set by the *spectral gap*. Sends the whole state every round.

Relaying (flooding / message passing)

Each round: forward fresher copies of *each agent's own* record (with a version stamp).

round	who knows "a_1 has 4"
1	a_2, a_4
2	everyone

Exact after D rounds (diameter); sends only records that changed.

UCB uses counts directly $(1/\sqrt{n_j})$: mixing gives *approximate* counts, relaying gives *exact but stale* ones.

Finding 2: relaying vs mixing (E15, E18, E20)

Design (torus, $N = 16$, same T_0)	**Regret ÷ server**	**Scalars sent**
Server + event-triggered sync	1.000	57k
Spanning tree + event-triggered flooding	1.030	60k
Flooding / consensus / push-sum every round	1.02	9.5M–295M

Link failures (ratio to server)	$q = 0.3$	$q = 0.6$	$q = 0.9$
Flooding + event flooding	1.06	1.09	1.27
Running consensus	1.04	1.10	1.50
Chebyshev (accelerated) gossip	2.1	3.5	7.3

- **Same regret, about 2000× fewer scalars:** consensus/push-sum need 140–285M scalars.
- On regret *alone*, mixing is not worse: consensus every round is as good or slightly better on well-connected graphs (E16).
- Relaying depends on the *diameter*; mixing on the *spectral gap*, and acceleration collapses when links fail.

When does relaying win? Scope and limitations

Our claim (not "relaying beats mixing" in general)

For nonparametric bandits whose sufficient statistic is a **sparse table** (bin counts and sums), relaying exact, event-triggered updates **matches the regret** of consensus-based protocols with **orders of magnitude fewer scalars**, and is more robust to sparse or unreliable graphs.

Limitations we state ourselves

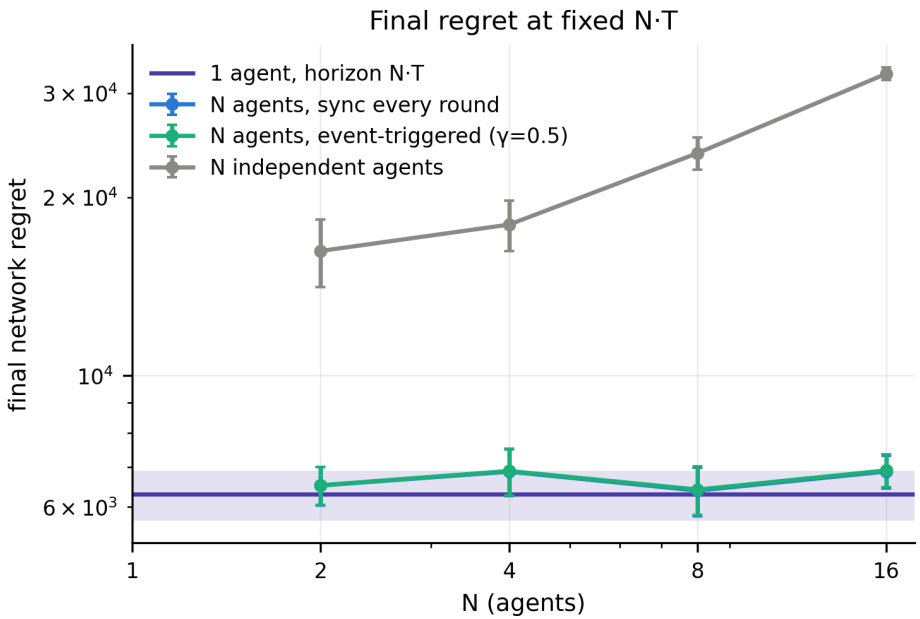
- Memory $O(N \cdot N_{\text{bins}})$ per agent
- Agents need unique IDs (no double counting)
- Individual agents' statistics are shared

Follow-ups that address them

- **Bounded memory:** relay *aggregated* tables along a spanning tree (as Phase 1 does)
- **Privacy:** agents add noise before relaying

Why most papers mix: constant memory in N, anonymous agents, built-in blending of data, a mature spectral-gap toolkit, and communication counted in *rounds*, where both cost one message per edge. **Precedent for relaying:** Dubey & Pentland (Coop-KernelUCB) use message passing for the same reason: averaging biases the estimates.

Reference: a server = one agent with NT pulls



E22: fixed budget $NT = 160$k, 20 paired trials (quadratic link)

- N synced agents: within **0–10%** of one agent at horizon NT, never better.
- Independent agents: **2.3–5.1**$\times$ worse.

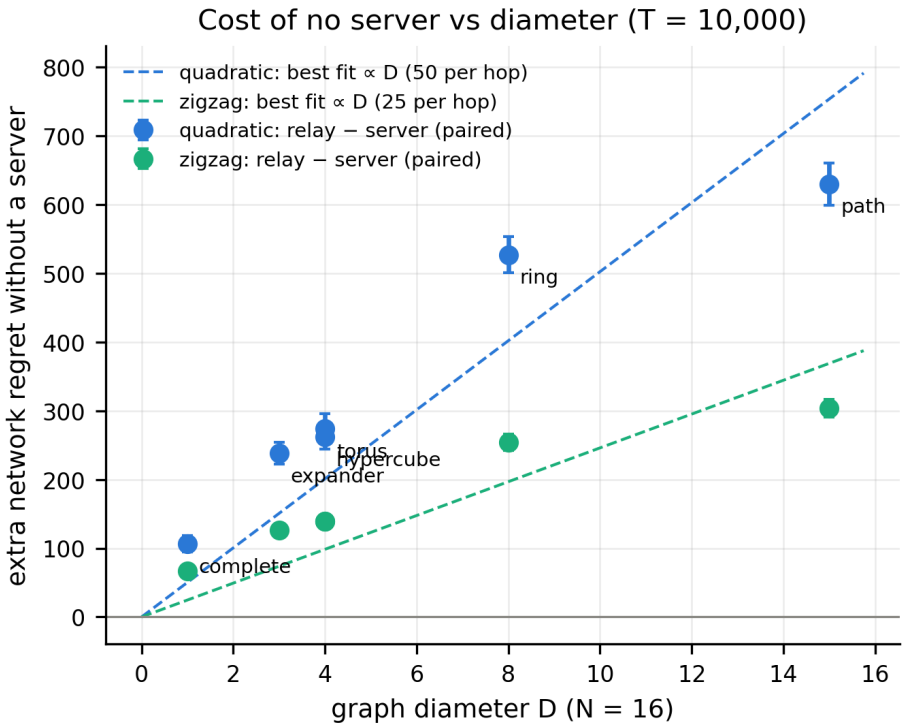
Proved (server)

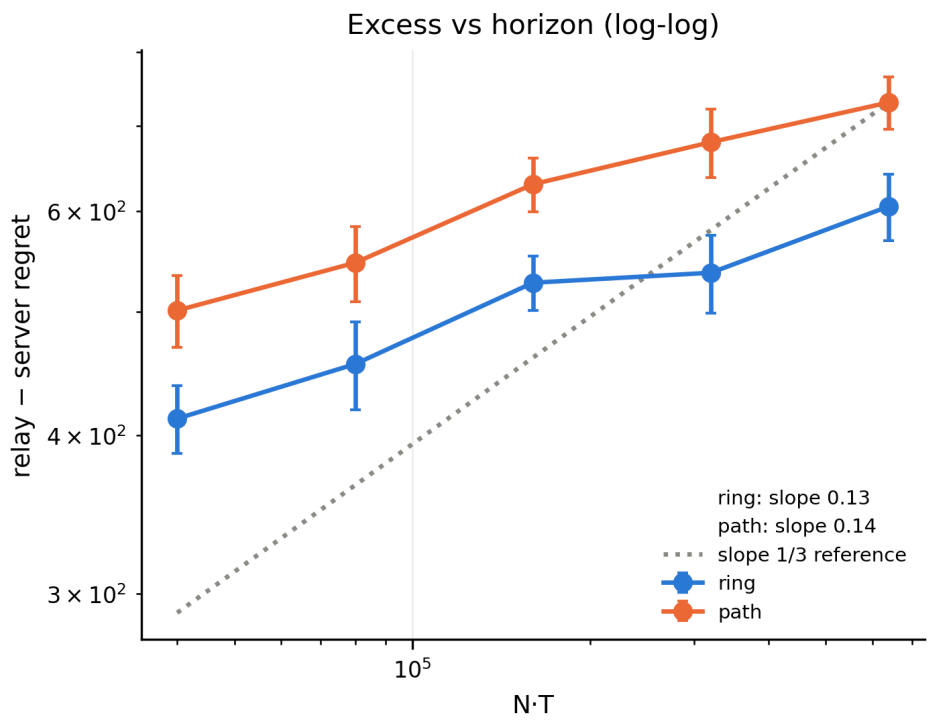
Lower bound by simulation: any N-agent algorithm $\geq c(NT)^{2/3}$.

Upper bound: Fed-ZoomSIB $= \tilde{O}\big((NT)^{2/3}\big)$.

The server rate is *closed*, so it is the yardstick for the serverless version.

Finding 3: what losing the server costs (E23)





- Extra regret: **+1.5%** (complete, $D = 1$) to **+9%** (path, $D = 15$), $N = 16$; grows *sublinearly* in D.
- Nearly flat in the horizon: log-log slope **0.13–0.14** against the $1/3$ our bound predicts.
- Relay sends 15–20% *fewer* scalars than the server (no broadcast to all N).

Theory: our bound vs the measurement

Theorem (relay, sketch)

$$R_T(N) = \tilde{O}\big((NT)^{2/3}\big) + \underbrace{O\big(N \cdot D \cdot N_{\text{bins}}\big)}_{\text{delay term}}, \qquad N_{\text{bins}} \sim (NT)^{1/3}$$

The gap

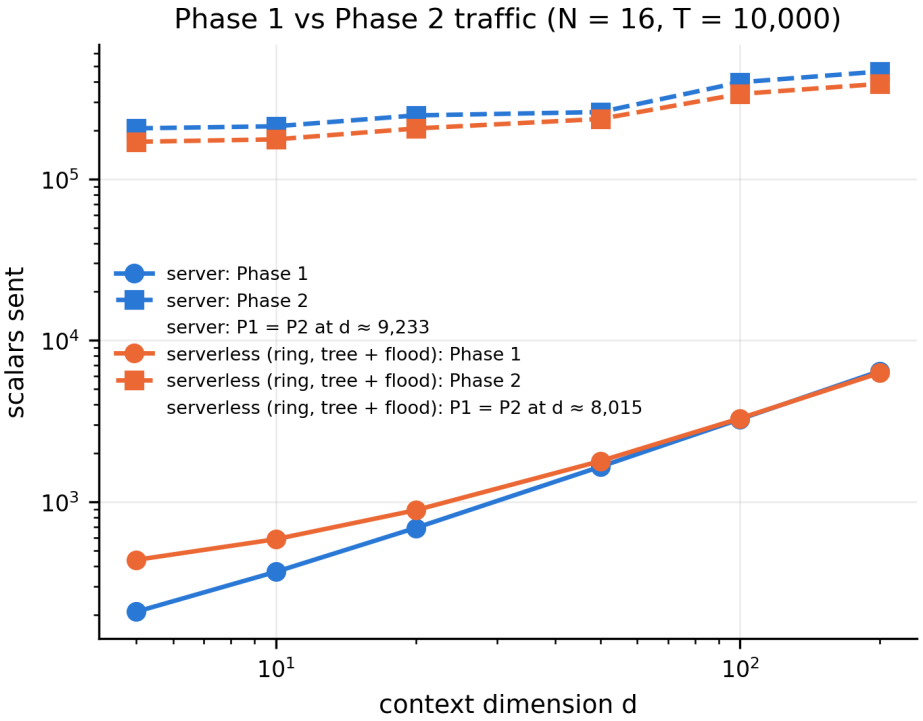
The delay term is **60–150$\times$ larger** than the measured cost, which barely grows with T.

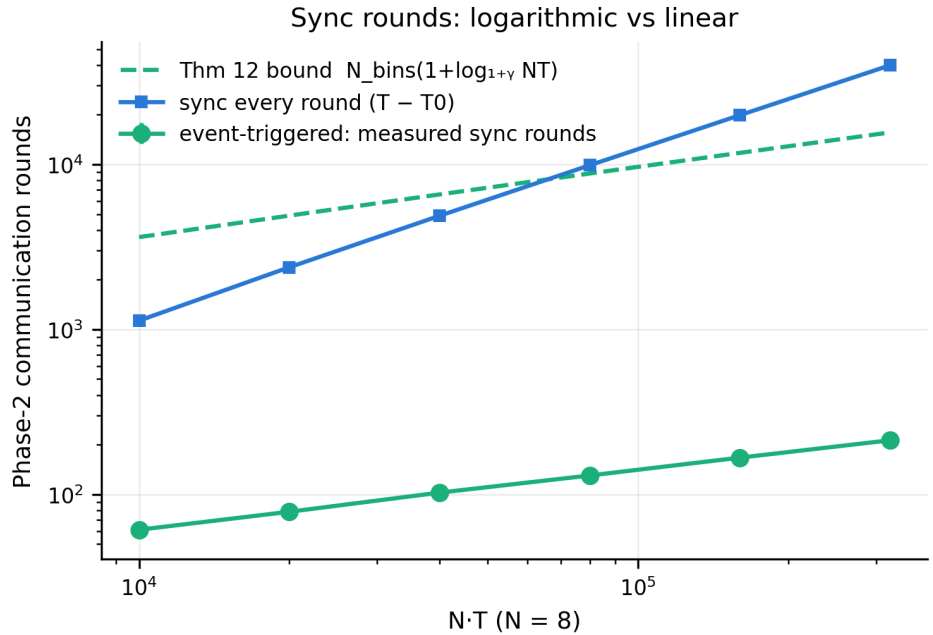
Prior art says it can be tighter

- DLUCB (decentralised linear): additive "regret of delay" only $\sim \log N / \log(1/|\lambda_2|)$.
- Coop-KernelUCB: the graph enters as $\sqrt{\bar{\chi}(G)}$ (clique cover).

Main theory target: only Phase-1 delay and *first visits* to bins pay the diameter.

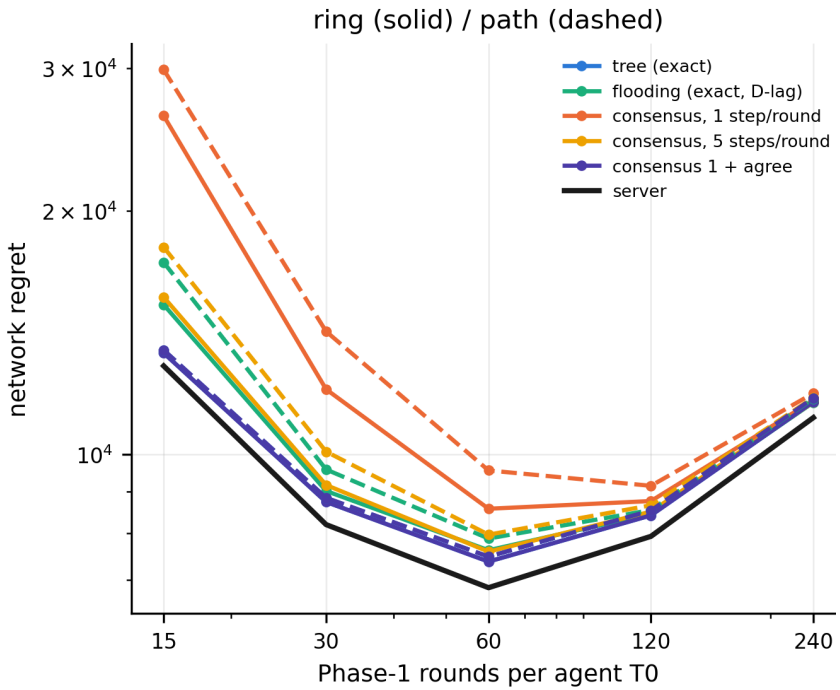
Finding 4: communication is all in Phase 2

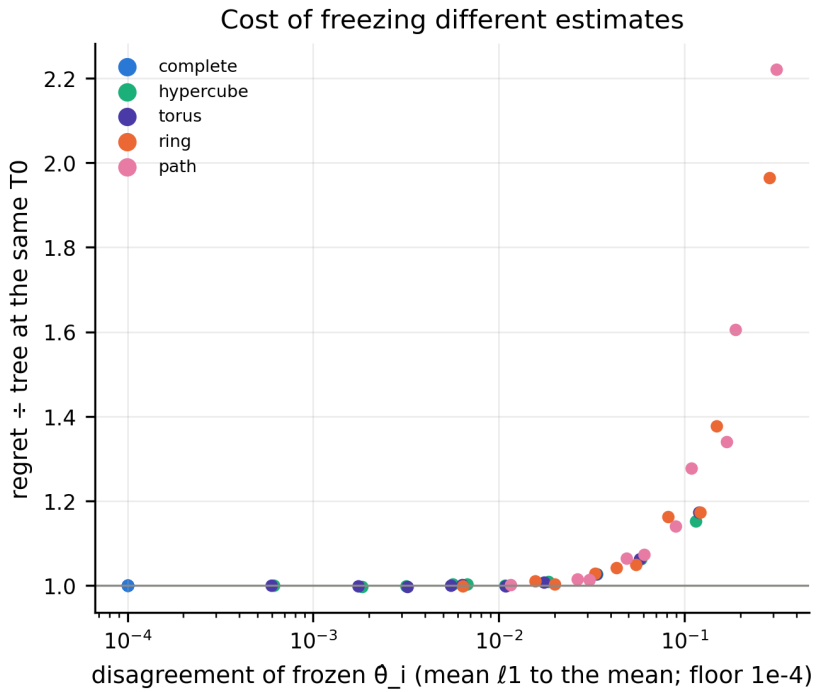




- Phase 1 is **0.1–1.6%** of all traffic ($d = 5 \ldots 200$), and shrinks with NT; compressing it saves $\leq 0.4\%$ of bits.
- Event-triggered sync: 61→213 rounds as NT grows $32\times$ (per-round sync: 1k→40k), at the same regret.

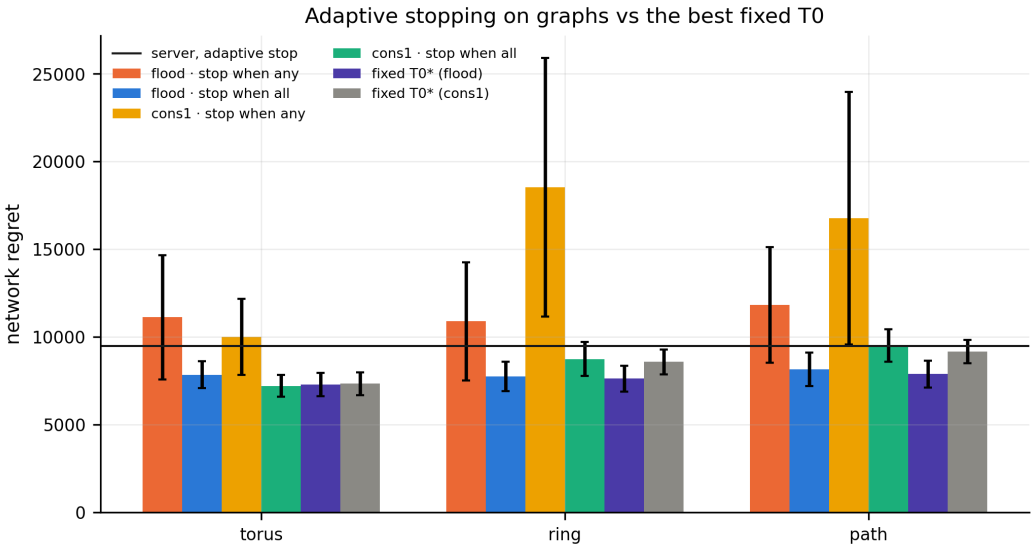
Finding 5: agents must agree, not explore longer





- The best exploration length T_0^* barely moves with the graph.
- The cost is agents freezing *different* $\hat{\theta}_i$: consensus costs $\times 1.25$ (ring), $\times 1.34$ (path).
- **One agreement pass** at the freeze removes it, for $\sim$2.8k extra scalars.

Finding 5: adaptive stopping on graphs fails



- "Stop when *any* agent is stable": **1.4–2.2**$\times$ regret, high variance.
- "Stop when *all* are": close to fixed T_0, but needs $\geq D$ extra rounds to detect (not modelled).
- Even the server's adaptive stop is worse than a fixed T_0.

Open question

A decentralised stopping rule that ends in *agreement*, with a regret guarantee.

Summary of findings

Claim	**Evidence**
Collaboration pays without a server	regret $\propto N^{0.44\text{–}0.55}$ vs $N^{0.98}$ alone (E19)
Relaying: same regret, far less talk	$\times 1.03$ of server at its communication; mixing needs $\sim 2000\times$ more scalars (E15, E20)
Robust to failing links	$+27\%$ at 90% link failures (E18)
Server $=$ one agent at NT	within 0–10%; proved lower $+$ upper bound (E22)
No server costs little	$+1.5\%$ to $+9\%$, nearly flat in T (E23)
Our delay bound is loose	60–150$\times$ above measurement (E23)
Communication is Phase 2	Phase 1 $< 2\%$; logarithmic sync rounds (E22, E24)
Agreement is what matters	disagreement $\times 1.25$–1.34, fixed by one pass (E28)

Baselines for the serverless setting

1. Reference baselines (as in DLUCB, Coop-KernelUCB)

No communication (independent ZoomSIB) and centralised (server) ✓ done

2. Prior papers' protocols, run with ZoomSIB

Running consensus (Landgren et al.), gossip (DDUCB), accelerated consensus (DLUCB), message passing (Coop-Kernel), push-sum ✓ done

3. Prior papers' algorithms, run on our problems

- DLUCB (decentralised linear): fails on non-monotone links ○ to do
- Dec-ESTOR / Dec-GSTOR: earlier single-index methods + our communication ○ to do
- Coop-KernelUCB (RBF): generic non-linear, pays in d and compute ○ to do

Next steps (Paper A)

Work	Deliverable	Status
Server theorem: lower + upper bound	proofs (`docs/proofs`)	✓
Relay theorem, full proof	Theorem with delay term	❍
Tight delay term (borrow DLUCB's technique)	additive, near-logarithmic cost	❍
Lower bound in the diameter D	matching necessity result	❍
Agreement lemma + dec. stopping rule	lemma + experiment	❍
Baselines: DLUCB, Dec-ESTOR/GSTOR, Coop-Kernel	experiment E29	❍
Literature check (multi-agent SIB)	related-work section	❍
Bounded-memory relay; private (noisy) relay	extensions	❍
Write-up	workshop / AISTATS-style draft	❍

Paper A: the first single-index bandit over a network: optimal with a server, and what the graph costs without one.

References

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Thank You!